

Programmability: T1 2D fabric vs SCAMP-5 pixel-processor array

SCAMP-5 has one analog processor per pixel but its ONLY links are the 4 nearest neighbours (N/E/W/S), run in with SIMD. Data transfer must moves must walk cell-by-cell; Non-uniform ops (each pixel a different distance from the target) need many shift rounds. T1 reaches any lane directly (gather / slide / H/V transpose) and reduction in hardware

**SCAMP-5 PE — 4 neighbours only**

data moves 1 cell per instruction

**1 lane — gather reaches ANY lane**

any source → any destination in same lane group in ONE instruction

Task A · move one pixel: top-left → top-right → bottom-right

**T1 fabric**

✓ 2 instructions

start → top-right → bottom-right

**SCAMP-5**

start → EAST → EAST → EAST → top-right → SOUTH → SOUTH → SOUTH → bottom-right

× 6 NEWS shifts

Task B · sum each row into the left column (vredsum) — an non-uniform move: every pixel is a different distance from the target

**T1 fabric**

✓ 1 instruction (every distance at once)

values → vredsum.vs → row-sums in col 0

**SCAMP-5**

values → shift 1 + add → after round 1 → shift 2 + add → row-sums in col 0

× log<sub>2</sub>W rounds, W-1 shifts (uniform only)

Summary

| Aspect                      | SCAMP-5                                                     | T1 2D fabric                                                                    |
|-----------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------|
| Inter-PE connectivity       | 4 nearest neighbours (N/E/W/S)                              | any pixel in same row/col → any pixel in same row/col (gather) + H/V transpose  |
| Relocate a pixel by (dx,dy) | dx + dy  sequential shifts                                  | 1 gather (vrgather)                                                             |
| Sum each row                | log <sub>2</sub> W shift-add rounds, all PEs shift together | 1 vredsum (hardware reduction)                                                  |
| Control granularity         | SIMD: every pixel moves the SAME way without masking        | High level lane process control, uniform & non-uniform, regular & irregular ops |